- PSD Functional Prototype and Accelerated Life Cycle Test
  - Project & Document Information
    - Project: MRT Orange Line (Bang Khun Non – Min Buri)
    - Subject: Test Procedure - PSD - Functional Prototype and Accelerated Life Cycle
    - Submission No.: PSD-40002.B
    - Date: 05-Aug-2025
    - Consortium: SML, SMOG, SMOA, SMT, STE (TH), BOZ
    - Test Location: ST Engineering Urban Solutions Ltd., Singapore
  - Test Overview & Requirements
    - Purpose: Ensure function meets design/contract requirements before production (Type Test)
      - Requirement: 1 Million Cycles (Continuous Open/Close)
      - Cycle Time: 12 seconds (6s Open; 6s Close)
      - Duration Estimate: ~4.6 months
      - Inspection Points: Start, 250k, 500k, 750k, 1M cycles
      - Monitoring: Surveillance CCTV for remote monitoring (Singapore)
      - Event Log: Record all stoppages/fatal alarms on HMI
    - Scope: Functional Prototype Test; Accelerated Life Cycle Test (Endurance/Wear)
    - Accelerated Life Cycle Test
    - Pass/Fail Criteria
      - Meet all requirements for each test case
      - Failures require investigation/analysis
    - Test Configuration/Equipment
      - Prototypes: One HHPSD doorway, One FHPSD doorway
      - Control Setup: PDP, PSC (Platform 1/2), DCU, LCP, DLCP, PTU
      - Simulator: Signalling System Simulator (Auto/Manual operation)
      - Measuring Tools: Oscilloscope, Linear Encoder, Force Gauge, Sound Level Meter
  - HHPSD Test Method and Requirements (Half Height)
    - Physical Check (ST-1000)
      - Visual Inspection: Dent/scratch check, cable support, sharp edges
      - Label Check: PSC, LCP, DLCP, DCU labels
      - Dimension Check: Frames (ASD, EED, FS), LCD Enclosure, Rear Cover
      - Coating Thickness Check: Min. 60µm
    - Electrical Check (ST-1010)
      - Insulation Test: > 5 kΩ (500VDC)
      - Continuity Test: < 1 Ω
    - Functional Tests
      - PSC Functional Test (ST-1110): Signalling Interface response (DOI, HMI status)
      - Door Operation Mode Test (ST-1120): LCP/DLCP manual control modes (Auto/Close/Manual Enable)
      - Door Motion Profile (ST-1130)
        - Closing Force: < 140N
        - Open Time: < 3.0s
        - Close Time: < 3.5s
        - Max Speed: < 0.5 m/s
        - Kinetic Energy: Max < 9.5J; Last 150mm < 2J
      - Obstacle Detection (ST-1140): Detect 10mm obstacle, 3 consecutive attempts (Nominal 3 cycles)
      - Portable Test Unit (PTU) (ST-1210): Verify software connection, modify parameters (re-try, reopen distance)
      - Acoustic Test (ST-1310): Max noise level 73 dBA (1m distance)
      - ERM Manual Release (ST-1410): Unlocking force ≤ 67 N, Opening force ≤ 133N
      - EED Test (ST-1510): Panic Bar force ≤ 67 N, Opening force ≤ 133N, Bypass functional
    - Maintenance Demonstration (ST-3000)
      - MTTR < 30 mins: DCU, Belt, Motor, Locking Block, Glass, ERM Housing, PLC, DLCP Switch
  - FHPSD Test Method and Requirements (Full Height)
    - Physical Check (ST-5000)
      - Visual Inspection: General fitting, sharp edges, coating color
      - Dimension Check: Frames (ASD, EED, FS), Anti-Climber, Vertical Gap Filler
      - Coating Thickness Check: Min. 60µm
    - Electrical Check (ST-5010)
      - Insulation Test: > 5 kΩ (500VDC)
      - Continuity Test: < 1 Ω
    - Functional Tests
      - PSC Functional Test (ST-5110)
      - Door Operation Mode Test (ST-5120)
      - Door Motion Profile (ST-5130)
        - Closing Force: < 140N
        - Open Time: < 3.0s
        - Close Time: < 3.5s
        - Max Speed: < 0.5 m/s
        - Kinetic Energy: Max < 9.5J; Last 150mm < 2J
      - Obstacle Detection (ST-5140): Detect 10mm obstacle, 3 consecutive attempts (Nominal 3 cycles)
      - Portable Test Unit (PTU) (ST-5210): Verify functionality and parameter modification
      - Acoustic Test (ST-5310): Max noise level 73 dBA (1m distance)
      - ERM Manual Release (ST-5410): Unlocking force ≤ 67 N, Opening force ≤ 133N
      - EED Test (ST-5510): Panic Bar force ≤ 67 N, Opening force ≤ 133N, Bypass functional
    - Maintenance Demonstration (ST-7000)
      - MTTR < 30 mins: DCU, Belt, Motor, Locking Block, Glass, ERM Housing, PLC, DLCP Switch